Cambridge Assessment International Education

# Cambridge International AS & A Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

*5395960624*

**PHYSICS** **9702/51**

Paper 5 Planning, Analysis and Evaluation **October/November 2021**

**1 hour 15 minutes**

You must answer on the question paper.

No additional materials are needed.

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

**INFORMATION**

- The total mark for this paper is 30.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **8** pages.


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**[Turn over**

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**1** A student investigates stationary sound waves in cylindrical tubes. Fig. 1.1 shows a stationary wave pattern in a tube which is open at both ends.

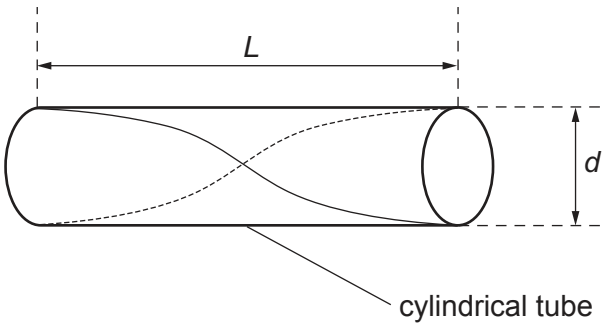



**Fig. 1.1**

The tube has length $L$ and diameter $d$. The frequency of the sound for the stationary wave pattern shown is $f$.

There are a number of different tubes available.

It is suggested that the relationship between $f$ and $d$ is

$$\frac{v}{f} = 2L + kd$$

where $v$ is the speed of sound in air and $k$ is a constant.

Design a laboratory experiment to test the relationship between $f$ and $d$.
Explain how your results could be used to determine values for $k$ and $v$.

You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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**Diagram**



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**2** A student investigates the discharge of a capacitor in the circuit shown in Fig. 2.1.

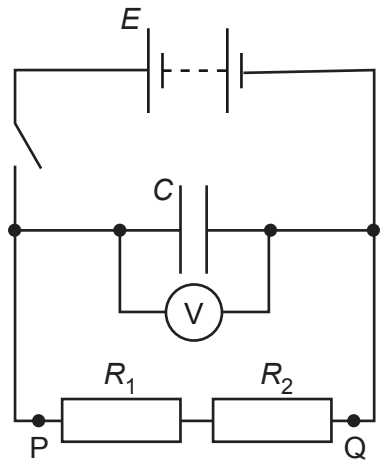



**Fig. 2.1**

The student closes the switch and charges the capacitor.

The switch is opened and a stop-watch is started. The capacitor discharges through the two resistors of resistance $R_1$ and $R_2$ connected between P and Q. At a fixed time $t$ the potential difference $V$ across the capacitor is measured.

The experiment is repeated for different values of $R_1$ and $R_2$.

It is suggested that $V$, $R_1$ and $R_2$ are related by the equation

$$\ln\left(\frac{V}{E}\right) = -\frac{t}{C(R_1 + R_2)}$$

where $E$ is the electromotive force (e.m.f.) of the battery and $C$ is the capacitance of the capacitor.

**(a)** A graph is plotted of $\ln V$ on the $y$-axis against $\frac{1}{R_1 + R_2}$ on the $x$-axis.

Determine expressions for the gradient and $y$-intercept.

gradient = .....

$y$-intercept = .....

[1]



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**(b)** Values of $R_1$, $R_2$, $V$ and $\ln V$ are given in Table 2.1.

Each resistance value has a percentage uncertainty of ±5%.

**Table 2.1**

| $R_1/\text{k}\Omega$ | $R_2/\text{k}\Omega$ | $(R_1 + R_2)/\text{k}\Omega$ | $\frac{1}{R_1 + R_2}/10^{-6}\,\Omega^{-1}$ | $V/\text{V}$ | $\ln(V/\text{V})$ |
|---|---|---|---|---|---|
| 22 | 33 | | | 1.28 | 0.247 |
| 22 | 47 | | | 1.98 | 0.683 |
| 22 | 68 | | | 2.87 | 1.054 |
| 33 | 47 | | | 2.39 | 0.871 |
| 33 | 68 | | | 3.28 | 1.188 |
| 47 | 68 | | | 3.55 | 1.267 |

Calculate and record values of $(R_1 + R_2)/\text{k}\Omega$ and $\frac{1}{R_1 + R_2}/10^{-6}\,\Omega^{-1}$ in Table 2.1.

Include the absolute uncertainties in $(R_1 + R_2)$ and $\frac{1}{R_1 + R_2}$. [2]

**(c)** **(i)** Plot a graph of $\ln(V/\text{V})$ against $\frac{1}{R_1 + R_2}/10^{-6}\,\Omega^{-1}$.

Include error bars for $\frac{1}{R_1 + R_2}$. [2]

**(ii)** Draw the straight line of best fit and a worst acceptable straight line on your graph. Both lines should be clearly labelled. [2]

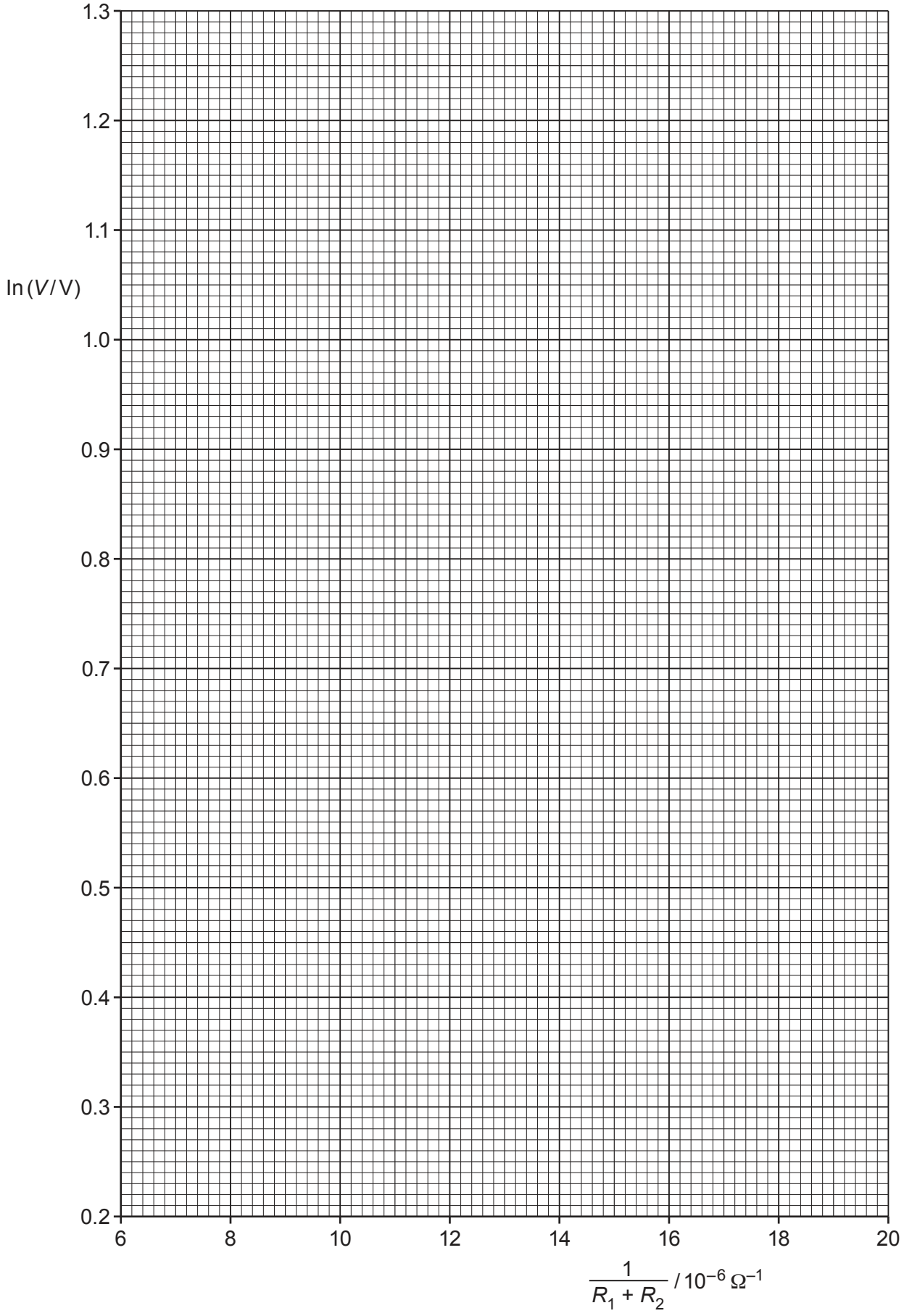
**(iii)** Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = ..... [2]

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**(iv)** Determine the *y*-intercept of the line of best fit. Include the absolute uncertainty in your answer.

*y*-intercept = ..... [2]

**(d) (i)** Using your answers to **(a)**, **(c)(iii)** and **(c)(iv)**, determine the values of *C* and *E*. Include appropriate units.

Data: $t = (60 \pm 1)\,s$

*C* = .....

*E* = .....
[2]

**(ii)** Determine the percentage uncertainty in *C*.

percentage uncertainty = ..... % [1]

**(e)** The experiment is repeated using the same capacitor. Determine the value of $(R_1 + R_2)$ that would give a value of *V* of 5.0 V at time $t = 60\,s$.

$(R_1 + R_2)$ = ..... Ω [1]

[Total: 15]

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